SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively.*(BL3)*

Assessment Tool Weightage: 35%

QUESTIONS: 3 **Duration : 1 Hour**
Total Marks:30

Annexure A

Constraint-Based Problem Solving

Code of Conduct:

I Vishwesh/192511383 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

VISHWESH K

Annexure A

Constraint-Based Problem Solving

1. A hospital needs to assign 3 doctors (D1, D2, D3) to 3 shifts (Morning, Afternoon, Night) such that:

   Constraints:

   i. D1 cannot work Night shift
   ii. D2 must work before D3 (Morning < Afternoon < Night)
   iii. Only one doctor per shift
   iv. D3 cannot work Morning
   v. All doctors must be assigned exactly one shift

   Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule

2. A robot operates in a grid environment (5×5). It must move from Start (S) to Goal (G). Obstacles block certain paths.

   Constraints:

   i. Movement allowed: Up, Down, Left, Right
   ii. Cost of each move = 1
   iii. Cannot pass through obstacles
   iv. Use Manhattan Distance heuristic

   Using above constraints and BFS Search Algorithm Show step-by-step node expansion and priority queue updates and determine the optimal path and cost**.**

3. An autonomous rescue robot is deployed in a disaster-affected building to locate and rescue survivors. The robot operates in a partially observable environment where some paths are blocked, and it must make decisions with limited information.

The building is represented as a grid of rooms. The robot starts at position S and must reach a survivor located at G.

Constraints:

- The robot can move up, down, left, right
- Some cells are blocked (obstacles) and cannot be traversed
- The robot has limited sensing capability (can only observe adjacent cells)
- Each movement has a cost of 1, but entering risky zones has an additional cost of 2
- The robot must minimize total cost while ensuring safe navigation
- The robot must update its knowledge dynamically (online search scenario)

Tasks:

a) Analyze all the given constraints and explain how they affect the robot's decision-making process.

b) Develop a solution approach using an appropriate search strategy (e.g., Uniform Cost Search / Online Search Agent). Clearly explain the steps involved.
c) Demonstrate how the robot applies AI concepts such as:

- Intelligent agents
- Rationality
- Environment type

d) Justify why your chosen approach is the most suitable for solving this problem under the given constraints.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Course Outcome: CO1 (AI Basics, Uninformed Search, Constraint Solving) **Total Marks:** 30

Assessment Weightage: 35% **Duration:** 1 Hour

Question 1: Constraint-Based Problem Solving (Hospital Shift Assignment)

Problem Statement: A hospital needs to assign 3 doctors (*D1, D2, D3*) to 3 shifts (*Morning, Afternoon, Night*) satisfying the following constraints:

- *D1* cannot work Night shift.
- *D2* must work before *D3* (*Morning* < *Afternoon* < *Night*).
- Only one doctor per shift.
- *D3* cannot work Morning.
- All doctors must be assigned exactly one shift.

Backtracking Search & Intermediate Steps

We apply Backtracking Search with constraint propagation to find a valid assignment:

1. **Assign Morning Shift:**
   - Options for Morning: *D1, D2, D3*.
   - Constraint check: *D3* cannot work Morning (Constraint iv). Thus, Morning can only be *D1* or *D2*.
   - **Try Morning = D1:** Remaining doctors for Afternoon and Night are *D2* and *D3*.
     - Next, assign Afternoon Shift from remaining *{D2, D3}*.
     - **Try Afternoon = D2:** Then Night Shift must be *D3*.
       - Check constraints:
         - *D1* (Morning) is not Night (Satisfied).
         - *D2* (Afternoon) works before *D3* (Night) → *Afternoon* < *Night* (Satisfied).
         - One doctor per shift (Satisfied).
         - *D3* is not Morning (Satisfied).
         - All doctors assigned exactly once (Satisfied).
       - **Valid Solution Found!**

> **Final Valid Schedule:**
> • **Morning Shift:** Dr. D1
> • **Afternoon Shift:** Dr. D2
> • **Night Shift:** Dr. D3

Question 2: Grid Navigation using BFS Search & Manhattan Distance

Problem Statement: A robot operates in a 5×5 grid environment, moving from Start (S) to Goal (G) avoiding obstacles. Move cost = 1 per step, using Manhattan Distance heuristic ($h(n) = |x_1 - x_2| + |y_1 - y_2|$).

Step-by-Node Expansion & Priority Queue Updates (Best-First / A* Search Approach)

Using informed search strategy (Best-First Search guided by Manhattan distance heuristic) to determine the optimal path and cost:

- Let Start $S = (0, 0)$ and Goal $G = (4, 4)$ in standard 0-indexed grid coordinates.
- **Initialization:** Open List contains $S(0,0)$ with heuristic $h(S) = |4-0| + |4-0| = 8$. Closed List = $\emptyset$.
- **Step 1:** Expand $S(0,0)$. Generate valid neighbors: $(0,1)$ with $h=7$ and $(1,0)$ with $h=7$.
  - Priority Queue (Open List): $[(0,1): 7, (1,0): 7]$. Closed List: $\{(0,0)\}$.
- **Step 2:** Expand $(0,1)$ (or $(1,0)$ due to equal priority). Assuming standard tie-breaking (Right then Down), expand $(0,1)$. Generate neighbors e.g., $(0,2)$ with $h=6$, $(1,1)$ with $h=6$.
  - Priority Queue: $[(1,0): 7, (0,2): 6, (1,1): 6]$.
- **Step 3:** Continuing expansions greedily towards the goal following minimal Manhattan distance while avoiding obstacles yields the optimal path along valid adjacent free cells.

> **Optimal Path & Cost Summary:**
> • **Optimal Path:** $(0,0) \rightarrow (1,0) \rightarrow (1,1) \rightarrow (2,1) \rightarrow (2,2) \rightarrow (3,2) \rightarrow (3,3) \rightarrow (4,3) \rightarrow (4,4)$ (or equivalent optimal grid trajectory)
> • **Total Optimal Cost:** 8 steps (Cost = $8 \times 1 = 8$).

Question 3: Autonomous Rescue Robot in a Disaster-Affected Building

Problem Statement: An autonomous rescue robot operates in a partially observable grid environment with obstacles, limited sensing (adjacent cells only), movement cost = 1, risky zone additional cost = 2, requiring online dynamic knowledge updates.

a) Constraint Analysis

Constraint	Impact on Decision-Making
Partial Observability & Limited Sensing	The robot cannot perceive the entire map upfront. It must rely on local sensing (adjacent cells) and update its internal belief state dynamically.
Obstacles (Blocked Cells)	Imposes hard boundaries. Unexplored blocked cells require runtime replanning when encountered.
Variable Costs (Normal = 1, Risky = 3)	Forces the pathfinding algorithm to balance safety and efficiency, avoiding risky zones unless the detour cost outweighs the risk penalty.

Online Search Requirement	Requires interleaving computation and action (e.g., Real-Time A* / Online DFS) since the environment is discovered incrementally.

b) Solution Development Approach

We deploy an **Online Search Agent using Real-Time A* (RTA*)** combined with local sensor mapping:

1. **Sense:** Read sensor data from adjacent cells to identify obstacles and risky zones. Update local cost map.
2. **Plan:** Compute heuristic-guided path estimates to the goal using modified edge costs ($c(u,v) = 1$ for normal, 3 for risky).
3. **Act:** Execute the immediate next best step.
4. **Update:** Learn obstacle/risk locations dynamically and refine heuristic values for visited states. Repeat until G is reached.

c) Application of AI Concepts

- **Intelligent Agents:** The rescue robot acts as an autonomous agent equipped with sensors (percepts) and actuators (movement actions) to achieve its rescue goal.
- **Rationality:** The robot maximizes its performance measure (minimizing time and risk, successfully reaching the survivor) given its current percept sequence and built-in knowledge.
- **Environment Type:** Partially observable, stochastic/dynamic (unknown obstacles discovered on the fly), sequential, and episodic/multi-step.

d) Justification

An **Online Search Strategy (such as RTA*)** is the most suitable approach because offline search algorithms (like standard A* or BFS) assume complete a priori knowledge of the environment. In a disaster-affected building where layout details and blockages are initially unknown and partially observable, the robot must dynamically adapt, perceive locally, and replan in real time to ensure safe and successful navigation.